

Sensitivity Analysis of Initial Conditions In Lagrangian Droplet train simulations.

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1 Introduction

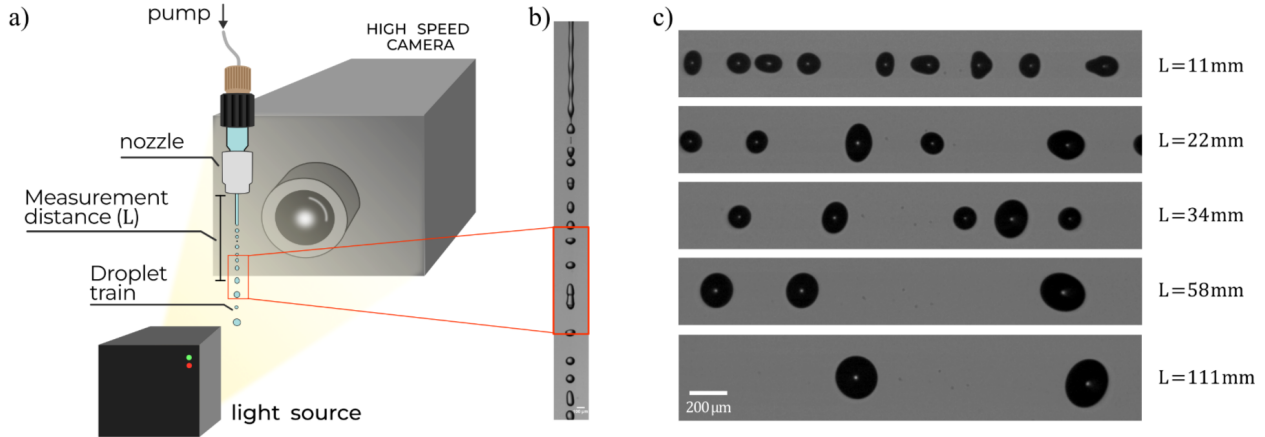


FIG. 1. Experimental configuration and spatial evolution of the spray. (a) High-speed imaging setup, with the camera aligned perpendicular to the jet axis and measurement planes from the nozzle to 111 mm downstream. (b) Near-field coalescence events where satellite droplets merge with main drops. (c) Far-field regime where droplet number decreases and characteristic size increases due to continued coalescence.

Figure 1: Figure taken from [1]

After the experiences of Stefan Kooji et. al.[1] (shown in figure 1) made in Amsterdam they find out that the droplet statistics after the Rayleigh-Plateau instability took the form of a Frachebourg distribution [2]. And their main conclusion is that droplets have a ballistic collisions and that everything can be explained by the ballistic theory [1] [3]. There for I have coded a Lagrangian droplet simulator that takes as input the droplet velocity distributions and the droplet diameter distributions and every time interval it selects a droplet to inject then takes a droplet diameter and velocity according to the distributions in the experiments that can be seen in the figure 2.

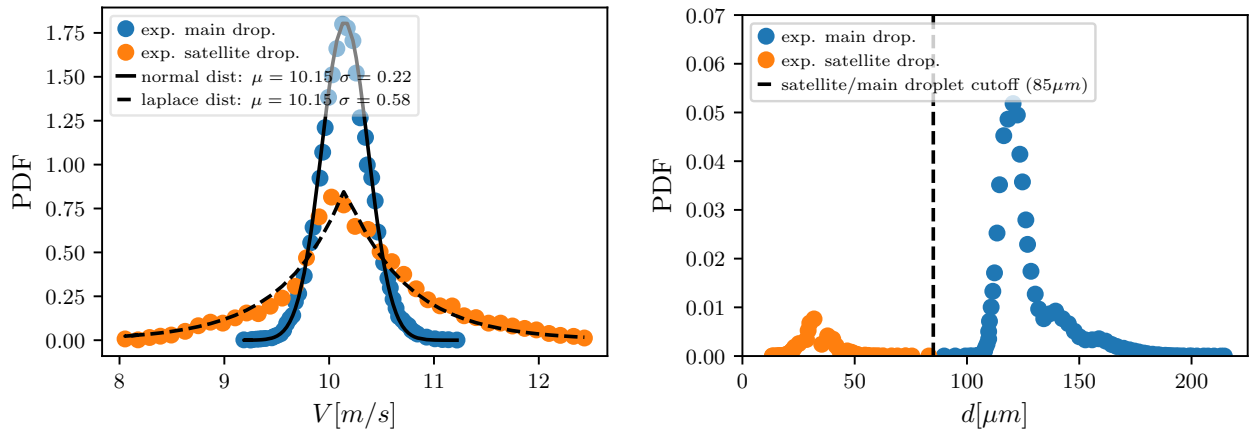


Figure 2: Experimental velocity(left) and diameter(right) probability distributions. Coloured by if the droplet is a satellite(orange) or main(blue) droplet. In lin-lin plotting.

2 How to run the program

The program is not the most polished program there for you might need some nuances to navigate through it. It uses the make compilation tool with a makefile. In that make file it

is important to put the correct paths MAT_NUM_INC and LAGJET_INC_DIR_SRC and there it need to be the directories corresponding to the Include paths of each respectively.

Then once everything is setup in the makefile inside the LagrangianJet directory you can run "make" in the terminal and it should compile the program. Once the program is compiled you can try to run the program in the command line using:

```
./exe.exe TestFolder/Standart dt=0.01 injection_rate=29.25  
recording_distance_in_mm=111 nb_droplets_saved=1000
```

Now normaly the program should run without any errors if there are any errors come see me please.

3 Tasks

1. Plot the volume weighted PDF for the 11mm and 111mm cases in the same figure one in lin-log and the other in lin-lin.
2. Convert the experimental volume weighted probability densities to non weighted probability densities. Both for the 11mm and 111mm. The x axis it the L111mmdia_x which is what diameter corresponds to the volume weighted PDF in L111mmdiaPDFByVolume_y. Likewise for the 11mm.
3. Plot the experimental non weighted PDFs that you found in the previous question.
4. From the theory find the Rayleigh-Plateau droplet formation frequency or time interval at 11mm from a jet that is started from a nozzle of diameter $64\mu m$ and a velocity of $10.15m/s$. (The answer is 29.25 but need the handwritten proof).
5. Now that we know our inputs run the program with experimental setup for 100000 droplets saved with an injection rate of $29.25\mu s$ and with a $dt = 0.01$ how does the distribution look like at 11mm and 111mm?

Bibliography

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- [2] L. Frachebourg, “Exact solution of the one-dimensional ballistic aggregation,” eng, *Physical review letters*, vol. 82, no. 7, pp. 1502–1505, 1999, ISSN: 0031-9007.
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